Enhancement Two Narrative

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CS-499: Computer Science Capstone

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1. Briefly describe the artifact. What is it? When was it created?

   The artifact is my AllCalculations.cpp and InputOutput.cpp files in my Investment Calculator Application that I created in CS-210 Programming Languages, back in February 2025.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in algorithms and data structure? How was the artifact improved?

   I selected the AllCalculations.cpp as my artifact for my second enhancement because it had some issues with how I was calculating the compounding interest with monthly deposits. The specific components of this artifact showcase showcases my skills and abilities in algorithms and data structures through how I enhanced my original two algorithms to be more efficient in the way it calculates the compounding interest for each year.

   I improved the algorithm by moving the calculations for monthly rate to the outside as more of a constant declared as a variable monthlyRate, which I was originally calculating it through each loop, which was unnecessary.

   I also selected the InputOutput.cpp again, because I needed to add a condition to the methods getInitialInfo() and menuSelection(). The enhancement came in the form of a condition for the amount of years so the user can't go over 100 years. Without this condition the user could type in any length of time in years and could cause the program to continue to run without stopping.

   I also took those same two methods and pulled the blocks of code that gathered the information from the use and created four more methods called getUserInvestmentAmount(), getUserMonthlyDeposit(), getUserInterestRate(), and getUserNumberYears(). This I never caught in my first enhancement and in my self-code review. The program used to have the same blocks of code in two different areas and now they just call them instead.

   I went through both files and removed unessasary inline comments and created comment blocks where needed. I also added headers to the top explaining the overall what the file does. I discovered an accident through reading an article on C++, where I was causing unnecessary flushes to the streams buffer by inserting new lines through 'endl', which I should have been using '\n' instead.

I also found unused variables in the AllCalculations.h file that I first thought I was using and discovered that I never once used them, the m_yearIndex, m_balance, m_interestEarnedThisYear.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

   I did meet the course outcomes that I planned to meet with this enhancement. I don't have any updates to my outcome-coverage plans.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

   As I went through the process of enhancing and modifying the artifact, I discovered something that I missed in my self-code review. I found that I was recreating the same blocks of code twice and for the same purpose but was using them in different areas. I made those blocks of code their own methods within the InputOutput.cpp file and just called them into the other methods that needed them. I discovered this during my testing phase where I made changes to conditions for user input and was still able to input what I was attempting to prevent.

   The main challenges that I faced was having to adjust my design from what I initially did in my enhancement one, without completely making the previous work pointless. I feel that I kept those enhancements from enhancement one intact but with some slight modifications the how and when.